REVIEW ARTICLE
https://doi.org/10.1038/s41590-017-0006-x

nature immunology

Corrected: Author Correction

The twilight of immunity: emerging concepts in aging of the immune system

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Immunosenescence is a series of age-related changes that affect the immune system and, with time, lead to increased vulnerability to infectious diseases. This Review addresses recent developments in the understanding of age-related changes that affect key components of immunity, including the effect of aging on cells of the (mostly adaptive) immune system, on soluble molecules that guide the maintenance and function of the immune system and on lymphoid organs that coordinate both the maintenance of lymphocytes and the initiation of immune responses. I further address the effect of the metagenome and exposome as key modifiers of immune-system aging and discuss a conceptual framework in which age-related changes in immunity might also affect the basic rules by which the immune system operates.

Aging leads to numerous changes that affect nearly every component of the immune system. Collectively called 'immunosenescence', these changes manifest themselves in increased morbidity and mortality of older organisms due to infectious disease[1]. While understanding of immunosenescence has progressed steadily over the past several decades, its underlying mechanisms are still not fully understood. Moreover, the interplay between the age-related changes that affect different components of the immune system remains incompletely elucidated, and there is no clear understanding of which changes are primary, arising as a consequence of aging, and which might be secondary, adaptive or compensatory to the primary changes. Gaining knowledge of these mechanisms and interactions will be critical for delaying the age-related decrease in immunity or preventing its consequences.

In considering how aging affects critical elements of the immune system (or any other organ or system), three distinct but interrelated components can be delineated: various immune cells; lymphoid organs; and circulating factors (chemokines, cytokines and other soluble molecules) that are not only produced by but also surround and guide the responses of both immune cells and lymphoid organs. Once that reductionist analysis is completed, which of the elements are most affected by aging and which may still be relatively intact can begin to be resolved. For the elements that are affected, the next set of questions should address whether the changes are intrinsic to that element or to the environment in which that element (typically a cell) operates. Following that, the goal is to obtain mechanistic insight into the basis of a given change. Finally, on the basis of the understanding described above, points at which intervention might be most effective to improve immunity should be determined. Notably, the most-effective interventions need not always be at the point of the most obvious 'lesion'. Instead, if that lesion is a secondary consequence of aging, the primary point of intervention may be upstream. For example, it has been shown that defects in T cell function can be effectively remedied by inflammatory cytokines and adjuvants[2,3]. This presumably improves the activation of T cells both directly and via the activation of dendritic cells (DCs). Indeed, adding pathogen-associated molecular patterns (PAMPs) to a subunit vaccine to trigger their corresponding pattern-recognition receptors (PRRs) has been shown to improve vaccine efficacy in older adults[4].

The complexity of the immune system mandates caution in studies of its aging. There are numerous cell subsets within each of the B cell, T cell, natural killer (NK) cell, DC, macrophage, neutrophil and other lineages, and aging often changes the relative abundance of such subsets. For example, it has long been known that naive T cells (T_N cells: $CD44^{lo}62L^{hi}$ in mice, and $CD28^{int}CCR7^{hi}CD95^{lo}CD45RA^{+}$ in humans) exhibit an absolute decrease with aging in both blood and secondary lymphoid organs (SLOs)[5]. That decrease is accompanied by a relative increase (and, in the case of infection with the ubiquitous pathogen cytomegalovirus (CMV), an absolute increase) in various populations of memory T cells (in particular, effector memory T cells (T_{EM} cells))[5]. Understanding true age-related changes within each subset critically requires the isolation, study and comparison of pure subsets. Otherwise, the results obtained from mixed populations have the potential to reflect the relative changes in subset abundance rather than the true age-related changes in a subset. In the example above, studying a proliferative response of total T cells will not provide information on whether there is a genuine defect in the proliferation of any of the T cell subset within the mixed population, rather than a mere shift in representation from a highly proliferative T_N cell population to a poorly proliferative T_{EM} cell population.

As mentioned above, the process of aging involves a combination of alterations in the following: (i) the cells of the immune system; (ii) the microenvironment in lymphoid organs and non-lymphoid tissues, where cells of the immune system reside; and (iii) the circulating factors that interact with both immune cells and their microenvironment to insure the proper initiation, maintenance and cessation of immune responses as well as homeostasis of the immune system. Much still remains to be learned about the three broad categories of changes noted above, particularly about age-related changes in the microenvironment and circulating factors, which are just beginning to be investigated. Some of these changes are not only quantitative but might in fact change the rules of operation of the older immune system, and that consideration should permeate thinking in this field. This Review will provide a summary of the better-understood aspects of 'immune aging' (the aging of cells of the adaptive immune system) and an overview of less-well-understood, emerging concepts related to coordination of immune responses, for which the potential for new and important knowledge is considerable.

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Age-related defects in innate immune cells

Since the early 1990s, the innate immune system has been increasingly recognized for both its critical role in overall immunity and its considerable complexity. In parallel, those in the field began to appreciate the extent of its alterations with age[6,7], although that knowledge still lags behind understanding of the aging of the adaptive immune system. An important issue to (re)emphasize here is that studies of the aging of the innate immune system are exceedingly sensitive to in vitro artifacts. First, the isolation of innate immune cells and molecules has the potential to pre-activate those cells and molecules and thereby distort their functional responses. Second, the in vitro population expansion of innate immune cells in the presence of cytokines and growth factors (for example, the in vitro differentiation of DCs and macrophages) has the potential to select for the 'best growers' and thereby obfuscate age-related differences that might exist in vivo. Next, as mentioned already, the diversity of innate immune cells also mandates careful separation of subsets (for example, M1 macrophages from M2 macrophages and NK cell subsets from one another) to avoid mixed and uneven population confounders. Finally, many innate immune cells of the same lineage can differ drastically in their longevity (DCs and macrophages in the blood and lymphoid organs live for a few weeks, whereas those in the tissues live for decades). Therefore, it is likely that short-lived subsets will be affected mostly by the aging of their environment or potential defects inherited from their precursor cells, whereas long-lived ones can accumulate true cell-autonomous, age-related defects. Thus, overall, it is imperative that studies of innate immunity be validated in vivo or be of minimally manipulated, ex vivo, isolated primary cells or molecules and, ideally, also use longevity tracing markers such as those that have been used for studies of T cells[8].

With that in mind, I will provide an extremely brief summary of the aging of innate immune cells, with the reader better served by more-extensive, topical reviews[7,9–11]. Globally, cytokine signaling, the production of peroxide and nitric oxide, and the phagocytic function of neutrophils (defined as $CD11b^+Gr\text{-}1^+$ cells) are all reduced in older people[6,7]. Such changes are often correlated with a poor prognosis in the face of bacterial infection and sepsis[12,13]. PRR signaling is reported to be diminished on old neutrophils, and in the case of the receptor TLR1, this might be linked to its lower expression[14]. It is unclear which of those defects are intrinsic to old neutrophils. NK cells exhibit age-related population changes, shifting from a less-mature, $CD14^+CD56^{dim}$, cytokine-producing subset to a mature, $CD14^+CD56^{bright}$ subset. Functionally, however, both subsets exhibit reduced activity (cytokine secretion and cytotoxicity) as well as diminished migration and altered diversity of activating and inhibitory receptors[11,15]. Such defects have been shown to directly increase the susceptibility of old mice to ectromelia[16].

The ability of macrophages to phagocytose apoptotic cells is somewhat diminished in both mice and humans[17], which raises the question of to what extent this impairment might fuel the low-grade inflammation discussed below. Defective responses to PAMPs have also been described[14,18,19], ranging from diminished phosphorylation of activating enzymes to decreased and delayed cytokine secretion and phagocytosis.

Age-related changes in DCs encompass reduced uptake of antigens and/or microbes and consequent diminished in vivo maturation, manifested by decreased migration and expression of costimulatory molecules and key cytokines necessary for T cell stimulation[20,21]. Moreover, decreased cross-presentation has also been described[22,23], and the importance of this defect has been particularly evident in cases in which microorganisms evade direct antigen presentation[24]. Clear defects in migratory properties have also been seen in certain DC subsets[20,25]. Similarly to old macrophages, old DCs exhibit diminished responsiveness to ligands of the innate immune system[18] at a transcriptome level and at a cytokine-secretion level. For both macrophages and DCs, it is controversial whether[14] or not[18] this might be due to the decrease in PRRs.

Overall, while innate immune cells exhibit numerous and clear defects with aging, it remains less apparent whether such defects (and which defects) are cell intrinsic or are patterned by the external environment.

Developmental and receptor-diversity changes with aging

T lymphocytes and B lymphocytes are substantially affected by the process of aging. Some of the changes occur early in the developmental progression from hematopoietic stem cells and are common to progenitor cells of both lineages. In addition, aging-related defects have been observed in the stroma of both the thymus and the bone marrow. Discussion of these changes surpasses the scope of this Review, and the reader is directed to other informative articles covering this topic[26–28].

At least for T cells, it appears that there is no major reduction in the diversity of T cell antigen receptors (TCRs). TCR diversity in older humans has been found to be reduced two- to fivefold in blood[29] and has been shown to be very diverse in the lymph nodes (LNs) of humans up to 60 years of age[30]. Data from mice are in agreement with that assessment (M.J. Smithey, personal communication). That stands in contrast with the mobilized TCR response in mice, which is severely restricted with aging[31].

Maintenance of lymphocytes in the periphery

Peripheral homeostatic mechanisms maintain the naive lymphocyte pool generated in youth for a long time[32]. However, eventually, age-related defects impair that maintenance[33]. The differentiation of naive T cells and B cells into memory lymphocyte subsets as a consequence of interactions with the microbial universe also occurs during the lifetime. This results in an absolute numerical depletion of naive T cells and B cells[34,35] that can also be seen at the level of individual antigen-specific precursor populations[36]. Naive B cells might suffer from an age-related decrease in BAFF (BlyS), a key maintenance factor for this lineage[37]. Instead, with aging there is accumulation of memory B cells that might be specific for environmental and/or self antigens and that carry a more restricted repertoire of B cell antigen receptors[37]. There is also an accumulation of unconventional aging-associated B cells that are independent of BAFF (BlyS) and that resemble marginal-zone and/or follicular B cells[38]. These cells also seem to respond 'preferentially' to endogenous antigens and are responsive to innate immune ligands, with potential role in autoreactivity[38].

For T_N cells, the key maintenance factor, IL-7, is not decreased with aging, but access to it in LNs is impaired, due to degradation of the LN architecture[39]. As this erodes the maintenance of true T_N cells (Fig. 1), some of the $CD8^+$ or $CD4^+$ T_N cells change their phenotype and convert into memory-like, 'virtual memory' T cells (T_{VM} cells)[36,40] (Fig. 1) that can perform various immune functions but that gradually lose the ability to fully expand their populations when confronted with microbial challenge[41]. It has been shown that the central memory T cell (T_{CM} cell) population in 18- to 22-month-old laboratory mice predominantly consists of such T_{VM} cells[42] and that these cells also exhibit a restricted TCR repertoire[41]. Therefore, both main lymphocyte subsets exhibit reduced maintenance of the naive subset and the accumulation of skewed memory subsets that are alternatively maintained, exhibit a restricted antigen-receptor repertoire and are probably both less able to undergo population expansion and function in immunological defense.

While aging of the adaptive immune system has been often described as a shift from naive lymphocytes to memory lymphocytes, that shift is only relative, and aging by itself does not lead to an absolute accumulation of memory T cells and B cells. Instead, an absolute increase in memory T cells is observed only in humans infected with CMV, particularly prominently as they age[35], and is a

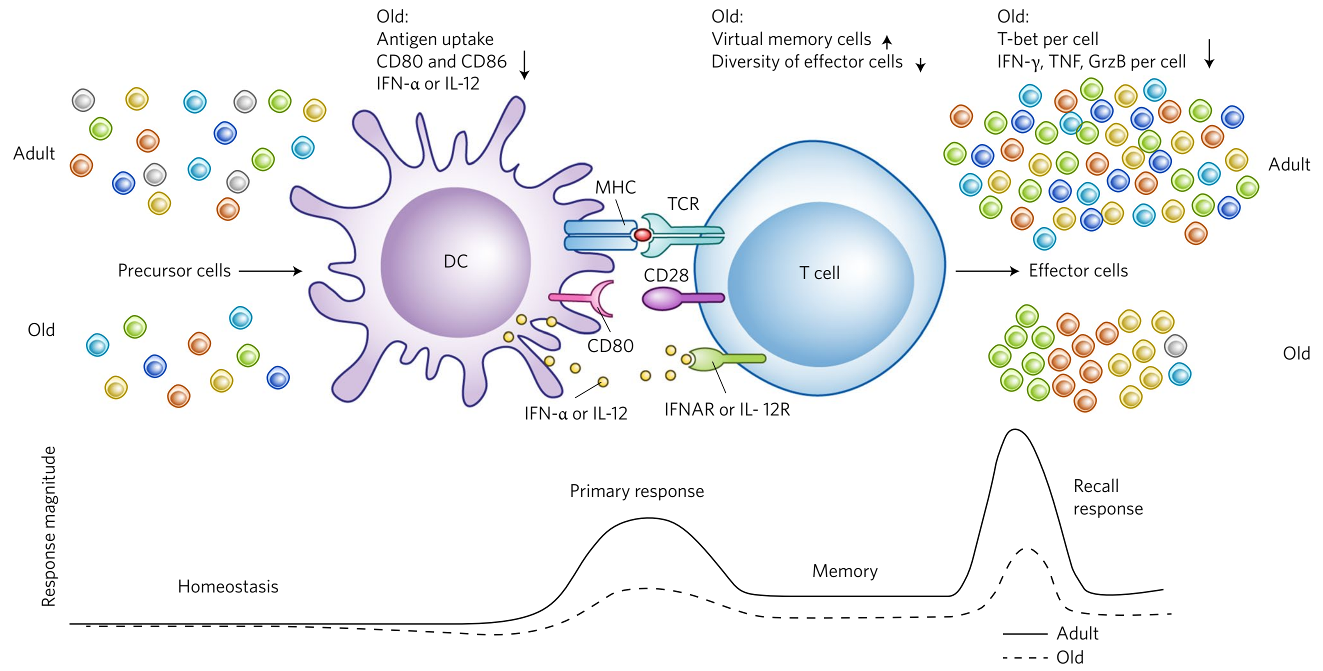

Fig. 1 | Defective activation of naive CD8+ T cell responses with aging. Aging results in a decrease in the number of naive CD8+ T cells, an increase in the number of memory-like precursor cells ('virtual memory cells') and a moderate reduction in the diversity of the TCR repertoire. During the primary response to intracellular pathogens in the aged environment, DCs exhibit impaired maturation, lower antigen uptake and reduced production of signal 3 (cytokines). Coordination of the response is also impaired (Figs. 2 and 3). The outcome is severely impaired population expansion of effector cells, diminished polyfunctionality and considerable narrowing and homogenization of the TCR repertoire elicited. Recall responses are similarly affected if the memory was formed in old age but are less affected if it was formed in youth. CD80 and CD86, co-stimulatory molecules; MHC, major histocompatibility complex; CD28, co-receptor; IFNAR, interferon-α receptor; IL-12R, IL-12 receptor; GrzB, granzyme B. Credit: Marina Corral Spence/Springer Nature.

consequence of the massive expansion of numerous clones of cells specific for CMV antigens[43–45], often also called 'memory inflation'[46] (discussed below in the section on the metagenome, defined as all the genetic material in a host that does not comprise the DNA of the host itself, including the genomes of many individual microorganisms).

Age-related changes in B cell functional responses

The functional responsiveness of B cells is believed to diminish with aging[37]. During activation, old B cells exhibit defects in induction of the key transcription factor E47 and consequent under-induction of AID, the key enzyme in charge of class switching and somatic hypermutation[47]. This leads to decreased avidity of antibodies in older people[48] associated with diminished antibody-mediated protection. Whether such defects are intrinsic to old B cells (and to which subset they are intrinsic) or are a consequence of impaired immigration into LNs[39,49] and/or reduced interactions with CD40L+ helper T cells[50] is less clear at this time.

The maintenance of memory B cells did not show obvious defects in an analysis of vaccination against influenza virus in hyperimmunized humans[48], but the authors observed a decrease in antibodies, suggestive of a defect in the differentiation of long-lived plasma cells (with reduced expression of the transcriptional repressor Blimp-1)[48]. In contrast, mice have defects in both memory B cells and the function of long-lived plasma cells (M.S. Diamond, personal communication).

Age-related changes in T cell functional responses

Original reports described reduced proliferation of T cells and production of IL-2 in response to mitogenic and anti-TCR agonist antibody stimulation[51,52]. However, those reports did not distinguish between the representation of naive lymphocytes and that of memory lymphocytes within the responding populations and therefore the results could be explained by a lower abundance of T_N cells and greater abundance of a poorly proliferative T_{EM} cell lineage within the old blood–spleen–LN population relative to their abundance in the adult samples. Indeed, experiments with sorted human cell subsets have rarely shown gross defects. A more subtle defect was identified in old human CD4+ T_N cells, whereby diminished TCR signaling and population expansion were linked to an age-related loss of miR-181, a key microRNA[53]. This microRNA normally represses the phosphatase DUSP4, which attenuates TCR signaling[53]. Our experiments with whole peripheral blood mononuclear cells and purified human CD8+ T_N cells suggest that proliferative defects that develop with age are visible only in mixed populations, whereas isolated and purified old CD8+ T_N cells exhibit few defects (V. Pulko, J.S. Davies and J.N.-Ž., unpublished observations).

CD4+ T cells from old wild-type mice or mice with transgenic TCR expression have been reported to exhibit TCR signaling defects during formation of the immune synapse and distal propagation of the signal[54–56], as well as in cytokine signaling[57,58] and lifespan regulation[59,60]. However, the authors of those studies analyzed either bulk wild-type cells (the caveats to which were discussed above) or CD4+ T cells from the transgenic mice, and they considered such transgenic cells naive by virtue of their lack of exposure to antigen and did not validate their actual naive-memory phenotype. The discovery of abundant populations of T_{VM} cells in old wild-type mice and mice with transgenic TCR expression[36,40,42] (Fig. 1) raises the possibility that the main defects were occurring in converted T_{VM} cells that accumulate with aging and not in true T_N cells.

In response to intracellular infection, mouse T_N cells exhibit defects in in vivo population expansion and differentiation, as measured by the induction of T-bet, the master regulator of the fate of the T_H1 subset of helper T cells, and downstream effector molecules, such as TNF, IFN-γ, granzyme B and others[61–63] (Fig. 1). Reductions have been found in both the frequency and the number of cells expressing those molecules, as well as in the amount of each molecule per cell[61–63]. Finally, while adult mice mobilize a highly diverse TCR β-chain variable region effector response with many clones and few overlaps between individual animals, old mice mobilize a highly restricted repertoire with two to four dominant clones taking up 70–80% of the repertoire and such clones are often shared by different animals[31] (Fig. 1).

For all of the changes noted above, it remains unclear whether they are true cell-intrinsic changes or whether a suboptimal priming environment, including reduced uptake and presentation of antigen[22,23,64] (Fig. 1), suboptimal chemotactic signals[49] and architectural changes in SLOs[39,49], also, or even dominantly, contribute to the defects observed.

It is possible that some of the changes noted above are mandated by epigenetic changes, which have been found to be pronounced with aging in several studies[65–67]. Indeed, one epigenetic analysis concluded that many T_N cells and T_{CM} cells assume a more-differentiated phenotype with aging[66], consistent with data from both mice and humans about conversion into memory-phenotype cells[36,40,42,68].

Aging-related signaling defects in populations of T_{EM} cells and memory T cells that re-express the naive-cell marker CD45RA but lack expression of the chemokine CCR7 (T_{EMRA} cells; collectively called '$T_{EM-EMRA}$ cells' here) have been a topic of several elegant studies of humans that have documented a block at the level of the mitogen-activated protein kinase p38 (ref. [69]) and the role of sestrins in that block[70] and have proposed means for its alleviation[69]. However, the larger teleological question here is whether such changes can actually be called 'age-related defects'. Specifically (as discussed below), persistent infections and, most notably, CMV expand large populations of $T_{EM-EMRA}$ cells, which can comprise up to 50% of the total T cell memory pool. From a homeostatic standpoint, after the next reactivation of a persistent infection with CMV, the last thing the organism needs would be another massive multiplication of virus-specific T cells. Therefore, having cytotoxic, cytokine-producing cells that control the infection but do not massively proliferate would seem to be the wisest strategy and therefore can be viewed as a beneficial adaptation, rather than a defect, of aging. From that standpoint, 'reversing' a proliferative 'defect' in highly differentiated $T_{EM-EMRA}$ cells might result in unwanted consequences for immune-system homeostasis. A separate issue is whether this population of cells could or should be partially depleted to reduce potential excess inflammation (if inflammation indeed originates from these cells) and how to achieve that.

Changes in soluble immunological mediators with aging

The most common description in the literature of the changes in soluble immunological mediators with aging is that of a low-grade persistent increase in inflammatory molecules[71], which has been called 'inflamm-aging'[72,73]. There are two problems with this description. First, neither that term nor any other term referring to elevated pro-inflammatory molecules in older organisms has been qualitatively or quantitatively defined. In humans, the phenomenon usually involves an increase in the level of IL-6, as well as somewhat more variable elevation in C-reactive protein, fibrinogen, IL-1β, TNF and other molecules[71]. But what level of each of these molecules indicates true 'inflammation' or a 'pro-inflammatory environment'? And how do elevated levels of several of these molecules combine to affect immunological and general aging? For example, is a 25% elevation in serum IL-6 together with a 40% elevation in IL-1β likely to result in a worse biological and clinical outcome than does a 50% increase in C-reactive protein and a 30% increase in serum TNF? At present, such information is sorely lacking. Currently, those in the field rely on clinical correlates that indicate to what extent an increase in an individual marker correlates with an increased risk of an adverse clinical outcome, such as cardiovascular events or all-cause mortality[74]. Furthermore, the source(s) of low-level pro-inflammatory changes with aging is (are) poorly understood. Candidates include increased adiposity with secretion of proinflammatory cytokines from the (altered) adipose tissue; increased translocation of microbes via an increasingly permeable gut barrier; accumulation and activation of highly differentiated T cells specific for persistent reactivating infections; and accumulation of non-lymphoid cells that assume the senescence-associated secretory phenotype[75,76].

Second, use of the term 'inflamm-aging' overlooks other concurrent and pervasive changes in soluble immunological mediators that occur in older organisms, both at a steady-state (homeostatic) level and in the course of the response to infection. The most cardinal and global change that affects older organs and tissues is fibrosis[73,77]. Fibrosis is a response to signals that lead to wound repair, and this response is dysregulated with aging[77]. A fibrotic response is under control of the cytokine TGFβ and of type 2 cytokines, in particular IL-13 (ref. [77]) (Fig. 2). An exuberant TGFβ response in older mice and/or humans has been described following systemic infection with *Encephalitozoon cuniculi*[78], chikungunya virus[63] or West Nile virus (J. Uhrlaub and J.N.-Ž., unpublished observations). In the case of chikungunya virus, neutralization of this cytokine in old mice was able to abolish the excess of age-related pathology[63], although the cellular source of the overproduced TGFβ and its effect on fibrosis were not elucidated. In another model, liver fibrosis was shown to be reversible by the chemokine CXCL9 (ref. [79]). However, in other experiments, increased TGFβ and IL-13 mRNA was detected in old LNs (H. Thompson and J.N.-Ž., unpublished observations), providing potential insight into age-related fibrosis in these organs. Overall, fibrotic changes occurring in response to injury (and aging) are characterized by increased thickness and deposition of collagen bundles in tissues and the proliferation and activation of myofibroblasts, which progresses to a manifest disorganization of tissue architecture (Fig. 2 and discussed below). This under-investigated pathology of aging broadly affects multiple organs, so that in the liver, initial steatosis (fatty liver) progresses to fibrosis, with a strong potential for cirrhosis and liver failure, as well as for hepatocellular carcinoma[80]. Overall, this set of global, age-related cytokine changes and the consequent fibrotic pathologies urgently require experimental elucidation and a considerable need for therapy.

SLO and tertiary-immune-site changes with aging

An underappreciated series of age-related changes have been discovered in old SLOs: LNs[33] and, to a lesser extent, the spleen. LN size decreases with aging, and one of the key clinical signs of infection, lymphadenopathy due to swelling of these organs, is known to be missing in older adults[33]. Over the past 20 years, it has become progressively clear that LNs represent not only the sites of initiation of many of the immune responses but also the key sites for the maintenance of naive lymphocytes[81,82]. Within the LNs, stromal elements (Fig. 3), such as two types of endothelial cells (lymphoid and blood), are in charge of lining the lymphoid and blood vessels, controlling antigen trafficking and producing numerous trophic factors that maintain neighboring cells[33]. Other cells, such as fibroblastic reticular cells, provide critical scaffolding and three-dimensional architecture to LNs, forming conduits that govern the trafficking of cells, in particular T cells[81]. Fibroblastic reticular cells also secrete IL-7 and manufacture extracellular matrix on which they deposit IL-7, presumably in a form accessible to T_N cells, and have been shown to be absolutely critical for the maintenance of T_N cells[82] (Fig. 3). Finally, LNs contain follicular DCs, which harbor soluble antigen in a 'depot' form and present it to B cells, as well as innate lymphoid cells, particularly group 3 innate lymphoid cells, that, in addition to their originally discovered role in LN tissue organization, might also have a role in LN maintenance[33].

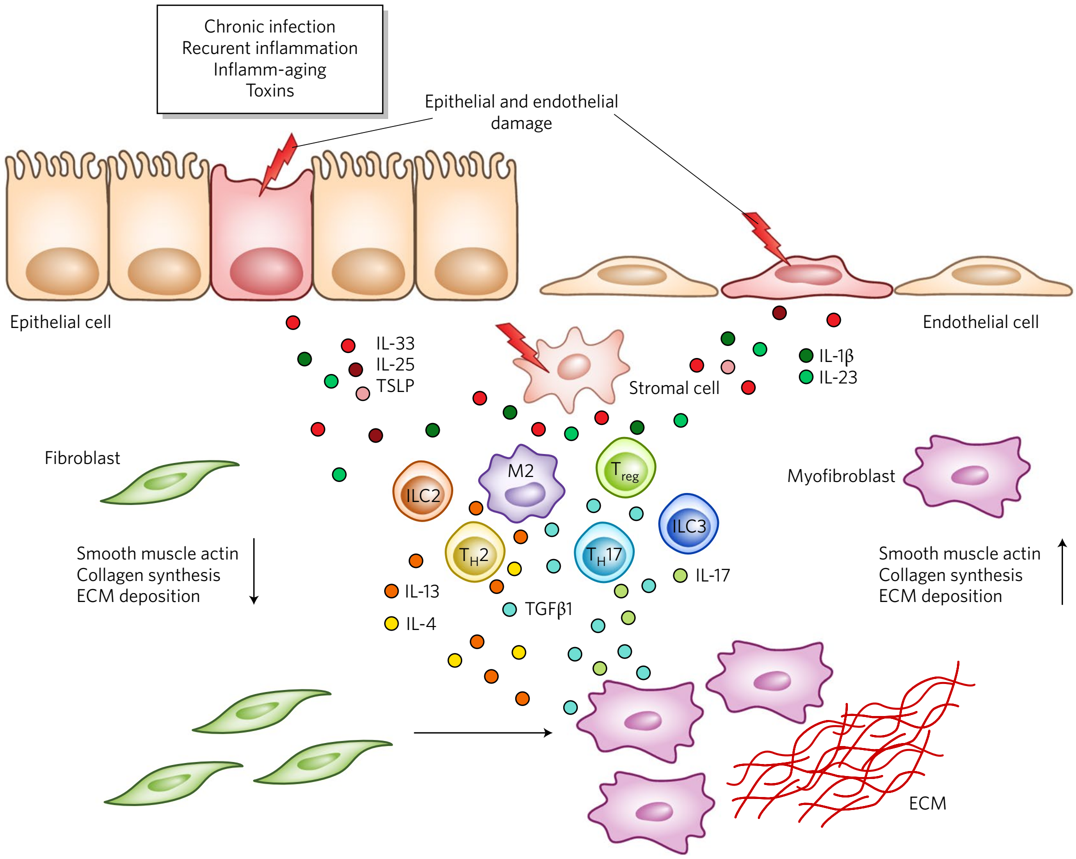

Fig. 2 | Fibrotic changes as a consequence of tissue damage and aging. Tissue injury or acute or chronic infection leads to the secretion of IL-33, IL-25 and thymic stromal lymphopoietin (TSLP) from epithelial cells and of IL-1β and IL-23 from endothelial cells. That activates a downstream cell (a T_H2 cell, regulatory T cell (T_{reg} cell), T_H17 cell, M2 macrophage or group 2 or group 3 innate lymphoid cell (ILC2 or ILC3)) to produce TGFβ1, IL-13, IL-4 and/or IL-17, which activate resting fibroblasts to become myofibroblasts and to increase secretion of collagen and smooth muscle actin and increase deposition of extracellular matrix (ECM). Credit: Marina Corral Spence/Springer Nature.

Several of the elements noted above, most notably fibroblastic reticular cells[39] and lymphoid endothelial cells (H. Thompson and J.N.-Ž, unpublished observations), have been found in some studies to decrease numerically with aging (Fig. 3), whereas other researchers have found no numerical changes[83]. Such differences probably arose from the fact that the last group of researchers included many CD45⁺ cells in their analysis, which potentially contaminated the stromal fraction with hematopoietic cells[83]. More importantly, structural studies have revealed a profound loss of architecture and obfuscation between discrete LN zones, so that T cell zones and B cell zones become blurry and overlap[39]. While the overall content of IL-7 mRNA does not decrease with aging, the proliferation of T_N cells in response to IL-7 does, and this can be remedied by exogenous injection of complexes of IL-7 and antibody to IL-7 (ref. [39]). Interestingly, in heterochronic parabiosis studies, old LNs did not increase in size when exposed to young blood elements and factors[84]. Instead, adult LNs shrank to the size of old LNs, and the most remarkable effect was noted for old stromal cells; that effect was reversed completely once the parabionts were separated, which strongly suggested that old blood contains a circulating cell or a factor that adversely affects the cellularity of LNs[84]. There is emerging evidence that old LNs exhibit fibrotic changes[33], with a potential to adversely affect both the homeostatic maintenance and immune responses of LNs. Consistent with that, cutaneous infection with West Nile virus[49] or chikungunya virus[63] has been found to lead to inferior swelling of old LNs relative to the swelling of adult LNs. In the first study, this was linked to poor recruitment of T cells and B cells to the affected LNs and to their slow and inefficient movement within the LNs during the course of the primary immune response[49] (Fig. 3). Transfer experiments have ascribed parts of this defect to T cells and B cells themselves, but a sizeable portion of the defect was found to be due to the aging of the LNs[49].

Elegant studies of humans have outlined discoordination of human T cell cutaneous immunity in old age. In response to ubiquitous antigens such as varicella-zoster virus, candida or mycobacterial purified protein derivative, older adults fail to mount a delayed-type hypersensitivity response despite having robust blood T cell responses to the same antigens[85]. Research has shown that antigen-specific T cells fail to migrate to the skin, due to defects in local macrophage production of TNF and consequent marking of blood vessel endothelium for recruitment and extravasation of T cells. However, macrophages retain the potential to secrete TNF, which suggests that this defect, too, results from mis-coordinated immunity at a local site[85].

All of the findings noted above suggest the possibility that structural changes in SLOs, in particular in LNs, as well as at tertiary immune sites, might represent an under-recognized key element in the deterioration of lymphocyte maintenance and function and mandate intense study of the underlying mechanisms to ameliorate this problem.

The metagenome and exposome

The immune system is modulated by many extraneous influences over a lifespan, including physical and chemical assaults (environmental

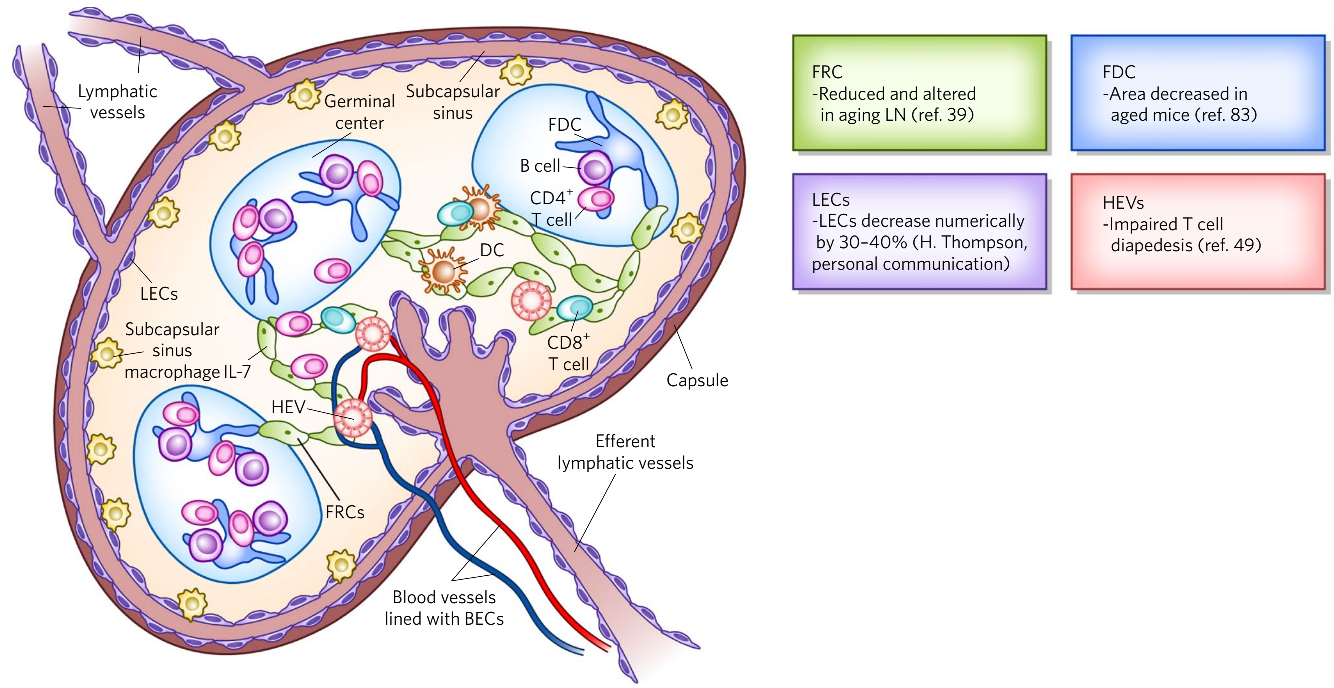

Fig. 3 | Age-associated changes in LNs. LN architecture and key elements (left): fibroblastic reticular cells (FRCs) make a mesh of conduits that facilitate the maintenance of T_N cells and trafficking of DCs and T_N cells, produce extracellular matrix, including collagen, and deposit IL-7 on them. Lymphatic endothelial cells (LECs) and blood endothelial cells (BECs) line lymphatic vessels and blood vessels, respectively, and these cells (particularly lymphoid endothelial cells) secrete various cytokines to maintain fibroblastic reticular cells. Follicular dendritic cells (FDC) serve as antigen depots for B cell responses. High endothelial venules (HEV) ensure the proper homing and extravasation of cells from blood to LNs, while group 3 innate lymphoid cells induce the formation of LNs and might maintain elements of LN architecture. Right, changes of the elements at left with aging (refs [39,49,83] and H. Thompson, personal communication). Credit: Marina Corral Spence/Springer Nature.

toxins, radiation, etc.) and exposure to a diverse world of microorganisms that the immune system must contain and/or eliminate. All of these factors, sometimes collectively called the 'exposome', have the potential to drastically affect aging of the immune system. Seminal experiments have shown that the last contact noted above is critical for the development of key parts of the immune system and other systems. For example, gut and gut-associated immunological tissues cannot develop properly in the absence of microbial colonization[86]. Similarly, microbial colonization is critical for determining the homeostatic proliferation of systemic T cells[87]. There is an accumulating literature on changes to the microbiome with aging, and in humans it has been shown that some of these changes correlate with the level of inflammation in older adults dwelling in nursing homes[88]. It has been hypothesized that the translocation of microbes across the gut epithelium might underlie this process, and while some evidence is starting to emerge[89,90], the evidence is not nearly as conclusive as it is for primary infection with human immunodeficiency virus[91]. Of interest, this mechanism could also provide new antigens that could stimulate subthreshold or manifest conversion of T_N cells into T_{VM} cells or T_{CM} cells. A published study has suggested that this mechanism might indeed be operative. One important note here is that in laboratory rodents kept under specific-pathogen-free conditions, the immune system remains underchallenged by environmental microorganisms and therefore resembles that of a neonatal human before full microbial colonization[92]. In contrast, adult wild-caught mice or mice from pet stores, as well as inbred specific-pathogen-free mice co-housed with 'cousins' from pet stores, exhibit a very different and much more adult-human-like immune system[92]. Better approximation of aging of the human immune system will require analysis of aging of the immune system in animals with lifelong exposure to a broad and diverse microbial environment more typical of human exposure. Similar pioneering studies with exposure of model organisms to air pollution typical of that experienced by humans have shown massive activation of the innate immune system that is very different from that seen in specific-pathogen-free animals[93]. All these experiments have the potential to shed light on critical aspects of immune-system aging.

In the context of immune-system aging and general aging, a special place within the metagenome belongs to persistent infections, which are not terminated to result in sterile immunity but instead remain to smolder for the life of the organism. Some of these infections, including infection with human immunodeficiency virus, hepatitis C virus or the parasite that causes malaria, affect a minority of the population (although the number of people infected can still be in the hundreds of millions), are chronic and systemic, and result in manifest clinical illness of the person. Unless they can be controlled by medication over the long term, they are not considered part of normal aging. In contrast, herpesvirus infections are ubiquitous, affecting over half of the total population, and, unless they reactivate in immunosuppressed people, rarely present a life-threatening health problem. Because the majority of humans age with these infections, they can be considered part of normal human aging. Such infections and their effect on aging are beyond the scope of this Review. I will discuss only the most prominent representative of this group, CMV, which is known to manipulate an enormous number of immunological parameters[94] and has been uniquely associated with immune-system aging, cardiovascular disease in older adults and even all-cause mortality[95]. CMV truly presents a 'yin–yang' face of immune-system aging. This virus drives an unprecedented overall antigen-specific population expansion of $CD8^+$ T cells and $CD4^+$ T cells over the course of a lifespan, with many of those being fully differentiated, highly cytotoxic and inflammatory-cytokine-secreting cells[43–45]. This provides a mechanism whereby 'trigger-happy' $T_{EM\text{-}EMRA}$ cells could secrete inflammatory cytokines

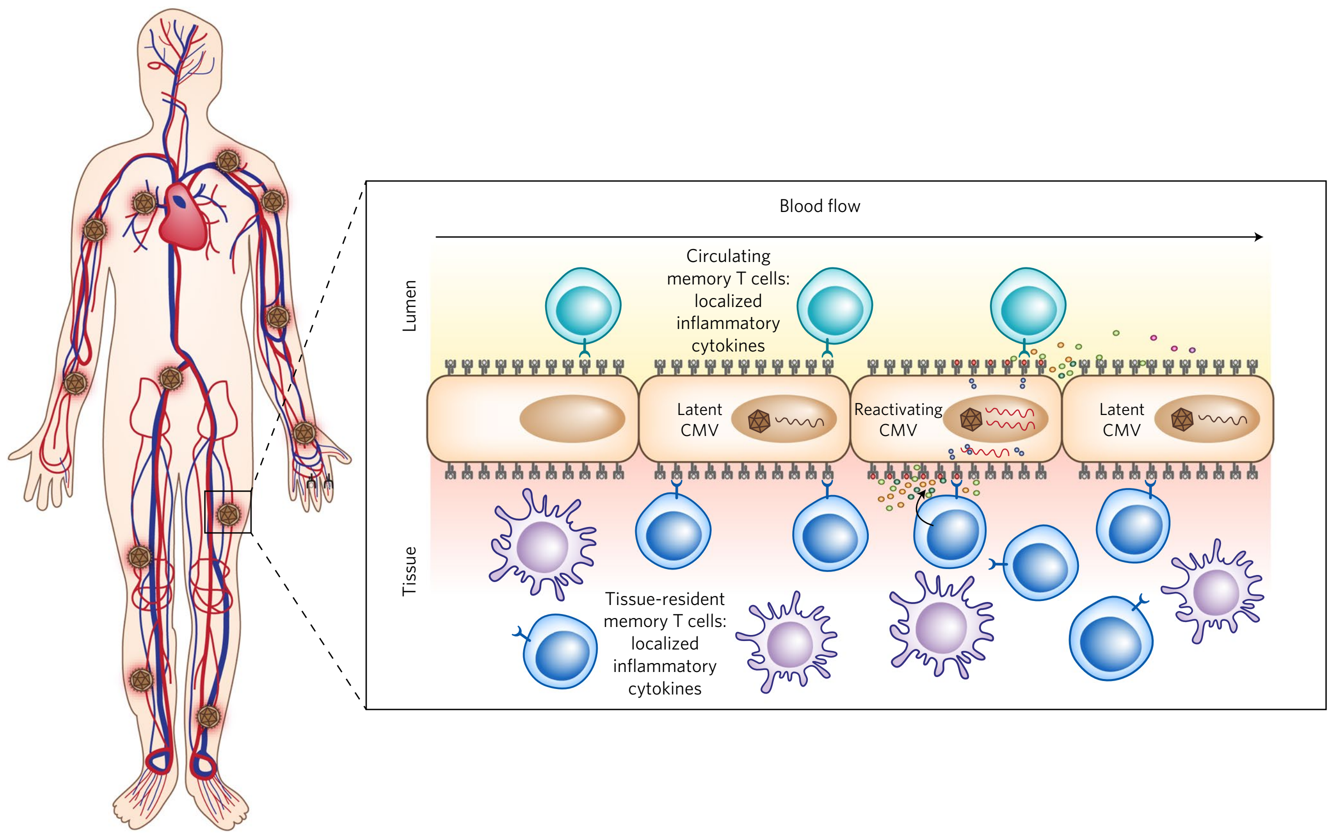

Fig. 4 | Systemic and tissue-specific consequences of CMV latency, micro-reactivation and full reactivation. CMV is distributed throughout the organism in cells of the monocyte lineage and endothelial cells. Reactivation events have been described as smoldering, discontinuous and affecting different cells in the body. Latently infected cells would produce latency-associated transcripts (maroon strand) that may or may not be antigenic. Those cells attempting to reactivate would produce lytic viral transcripts (red), which results in antigen presentation (red peptides) by the major histocompatibility complex molecules on the host-cell surface. That would lead to the recognition of antigen by both circulating CMV-specific T cells and resident-memory CMV-specific T cells, which would lead to their release of effector cytokines targeted to the reactivating cell. The majority of those cytokines would remain local or would be released into circulation at such a low level that they would not be detected by serum analysis. This would be the most common scenario in a healthy, immunocompetent host. Full viral reactivation would occur only in an immunocompromised host, with massive activation of circulating and resident-memory CMV-specific T cells and robust secretion of proinflammatory effector cytokines detectable by serum analysis.

and cytolytic molecules that could both increase overall inflammation and damage bystander cells and tissues. CMV also infects cells of the monocytic lineage, as well as endothelial cells and potentially other cells (Fig. 4), which explains why it is often found at sites of arterial disease[96]. At present it is unclear whether CMV is predominantly under the control of recirculating blood CD8$^+$ T cells or tissue-resident CD8$^+$ T cells. It has been postulated that the population expansion of T cells induced by CMV might constrict the remainder of the T cell repertoire and infringe on immune responses to other microorganisms. Indeed, studies of mice with lifelong infection with CMV have found evidence of somewhat diminished immunity[97–99]. However, there is no evidence of increased mortality from other infections in CMV-infected animals[97–100]. Moreover, there is a growing literature indicating that CMV can be useful to an immunocompetent host. Studies have found improved heterologous immunity in mice persistently infected with herpesvirus, via increasing activation of innate immunity[101], and have found improved responses to vaccination in adult CMV-infected humans but not older CMV-infected humans[102]. Notably, studies have so far failed to account for viral activity in different hosts to determine whether individual control of CMV latency and reactivation[103] might help to better explain its effect on immune-system aging. Moreover, laboratory mouse models also suffer from a lack of natural stressors that would physiologically reactivate the virus in the host (psychosocial, physical, chemical and infectious stressors). Both of these shortcomings are being addressed in ongoing studies that will hopefully substantially advance the field.

Infection with human immunodeficiency virus represents a special case of chronic infection that has been rendered survivable by successful antiretroviral therapy. In the Western Hemisphere, this infection is no longer lethal itself, although aging while infected with this virus carries a special set of problems, including early comorbidities and progeroid phenomena[104,105]. There is intense research on this topic to elucidate to what extent the virus, virus-induced gut-barrier damage, antiretroviral therapy and concurrent opportunistic infections all contribute to these sequelae.

Whether and how the metagenome and/or exposome interact(s) with cellular aging to produce epigenetic changes in lymphocytes is unclear at present.

Conclusion: changing rules and future perspectives

While the picture presented above has indicated numerous defects in the old immune system, many of which clearly precipitate poor defense against infection, there are reasons to remain optimistic about delaying or improving age-related immunological defects. First, aging is a plastic process that can be modified by nutritional intervention as well as pharmacological intervention[106,107]. Aging of the immune system is similarly plastic. It has been shown to be malleable by improved activation of both the innate immune system and adaptive immune system[2,4], by alteration of homeostasis via the manipulation of cytokines (neutralization or blockade or stimulation with cytokines or cytokine–antibody complexes)[39], by alteration of metabolic pathways in innate and adaptive cells[108–110], and by immune-system rejuvenation aimed at reigniting the production of new naive lymphocytes[111]. Targeting lifespan and 'healthspan' (the length of a lifespan spent in good health) with drugs to achieve partial inhibition of the metabolic checkpoint kinase and rapamycin target TOR has shown substantial promise. Multiple clinical trials are recruiting subjects for or have recently completed analysis of the immunosuppressant rapamycin (sirolimus) in elderly volunteers for effects on metabolism, muscle, skin, heart and breast cancer (information available at http://clinicaltrials.gov, with the search terms Aging + Rapamycin or Aging + Sirolimus). However, in the context of the immune system, even low doses of rapamycin impair the generation of effector T cells in (old) mice[112,113]. A modest improvement in antibody titers (but not T cell function) has been reported in subjects treated with rapamycin analogs, rested (washout period) and then vaccinated with a vaccine directed against seasonal influenza virus[114]. However, neither immunity nor vaccine efficacy during treatment with a low dose of sirolimus or other treatment with rapamycin analogs has been assessed in humans. Overall, as effector immunity deteriorates during aging, caution is mandated for the use of compounds that are known to or are likely to inhibit the population expansion and differentiation of naive lymphocytes into effector cells.

Going forward, it will be important to conceptually reassess how aging of the immune system is approached, because some of the changes that occur in the old immune system might not only erode certain aspects of immune responses but also fundamentally change the rules of operation of the old immune system. For example, the alterations in the structure of LNs discussed above would change the timing and type of engagement between T_N cells and their supporting structures, as well as between T_N cells and DCs. Similarly, the presence of large amounts of TGFβ would deviate immunity toward type 2 and pro-fibrotic responses.

That being said, major advances can be expected in the near future in areas of SLO aging, in research on soluble pro-geronic and anti-geronic factors discovered via heterochronic parabiosis, in elucidation of the interplay between persistent infections (and, specifically, CMV latency and reactivation) and the old immune system, in studies of aging mice exposed to a diverse microbiome, and in high-resolution studies of aging of the human immune system. Another exciting and poorly explored area is cancer immunotherapy in older people, which is critically important given that cancer predominantly affects the population over 50 years of age. In this context, T cell exhaustion does not seem to be a major mechanism linked to aging of the immune system, but whether or not that is the case for antitumor responses remains to be seen. Finally, there will probably be exciting translation of some of these findings into treatments and vaccines to benefit older adults, as has been shown with the new subunit vaccine against herpes zoster, adjuvanted with PAMPs[4].

Received: 2 October 2017; Accepted: 26 October 2017;
Published online: 14 December 2017

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Acknowledgements

I thank past and present members of the Nikolich lab and the UA Department of Immunobiology for collaborative work that led to some of the concepts crystallized in this work; E. Goldberg for suggestions; I. Jeftic, M. Jergovic, M. Smithey and H. Thompson for help with illustrations and critical perusal of the manuscript; and M. Kuhns for the suggestion that rules of immune system might change with aging and for critical perusal of the manuscript. Supported by the US Public Health Service (AG020719, AG048021 and AG053259), the US National Institutes of Health (HHSN 272201100017C and HHSN272200900059C) and the Elizabeth Bowman Endowed Professorship in Medical Science.

Competing interests

The author declares no competing financial interests.

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